

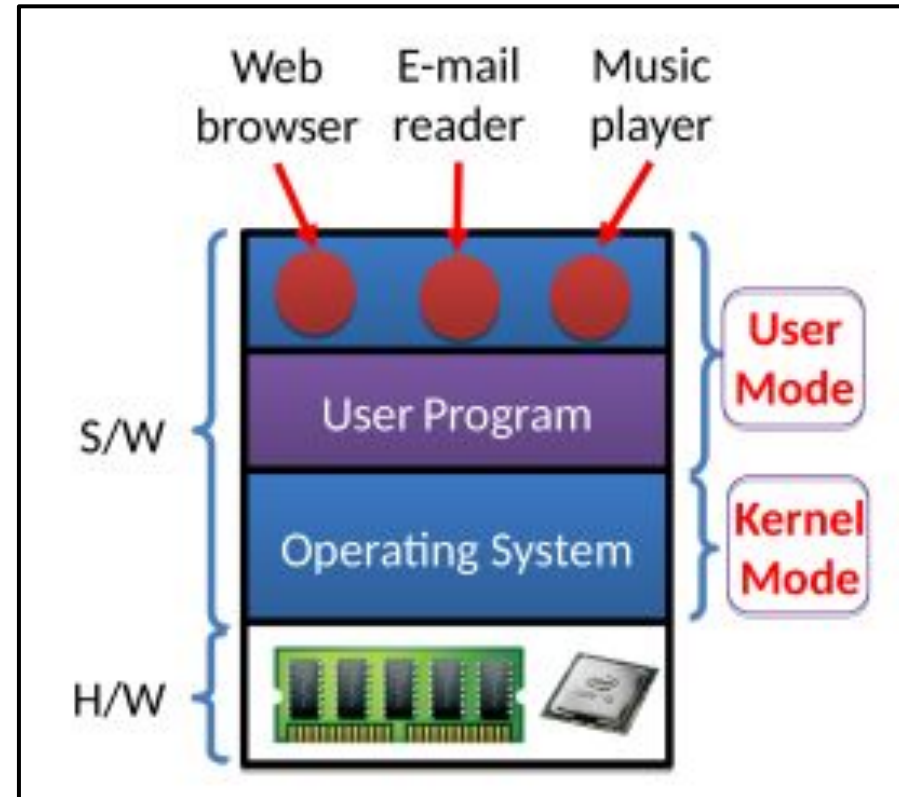
Unit –I Introduction To Operating System

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1. ☐ Meaning of OS
2. Functions of OS
3. OS Types
4. ☐ Introduction of OS process
5. ☐ Process State Transition Diagram
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Meaning of OS

- An operating system is like a bridge between your computer's hardware and the programs you use. It makes sure that your computer's memory, CPU, and storage are used effectively while running. The OS helps programs talk to the hardware, like managing how they use memory or control devices.
- Kernel mode: has complete access to all the hardware can execute any instruction that a machine is capable of executing has high privileged (rights)
- User mode: can execute only subset (few) of the machine instructions has less privileged (rights)

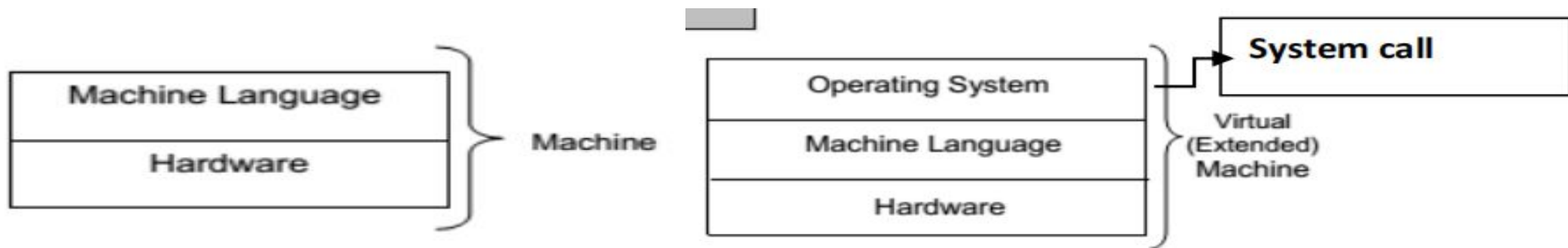


Operating systems perform two basically unrelated functions:

- Give the view of OS as an extended machine
- Give the view of OS as a Resource Manager
 - Time multiplexing
 - Space multiplexing

Give the view of OS as an extended machine

- The major task is to convert human readable form in to machine level language vice versa
- Users do not want to be involved in programming of storage devices and such kind of converter.
- Operating System provides a set of basic commands or instructions to perform various operations such as read, write, modify, save or close.
- Dealing with them is easier than directly dealing with hardware.
- To deal with such complexity, system program is available known as *system call*.
- Thus, Operating System hides the complexity of hardware and presents a beautiful interface to the users.
- The operating system provides a variety of services that programs can obtain using special instructions called system calls.



Give the view of OS as a Resource Manager

- A computer consists of a set of resources such as processors, memories, timers, disks, printers and many others.
- Multiple user programs are running at the same time.
- The processor itself is a resource and the Operating System decides how much processor time should be given for the execution of a particular user program.
- The primary task of OS is to keep the track of which programs are using which resources, to grant resource requests, to account for usage, and to resolve conflicting requests from different programs and users.
- An Operating System is a control program. A control program controls the execution of user programs to prevent errors and improper use of computer

Give the view of OS as a Resource Manager

- **Time multiplexing**

When a resource is time multiplexed, different programs or users take turns using it. First one of them gets to use the resource, then another, and so on the algorithm known as time multiplexing. For example, CPU and printer are time multiplexed resources. OS decides who will use it and for how long.

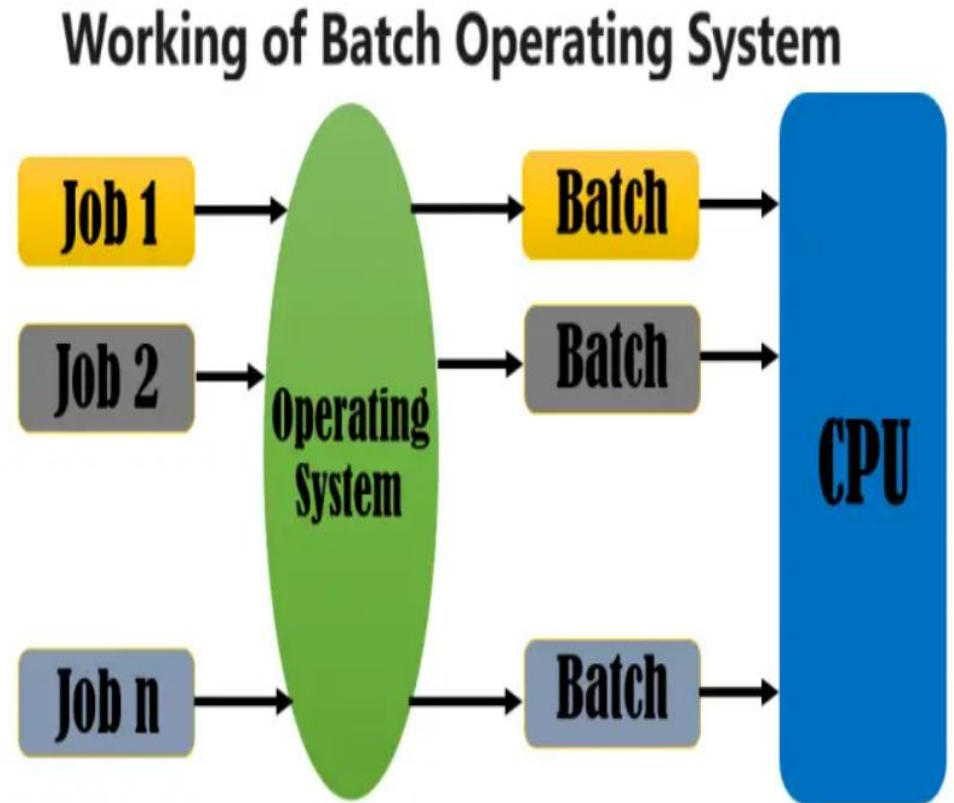
- **Space multiplexing**

The other kind of multiplexing is space multiplexing, instead of the customers taking turns, each one gets part of the resource the algorithm known as space multiplexing. For example, both primary and secondary memories are space multiplexed. OS allocates them to user programs and keeps the track of it.

1. Batch Operating System
2. Time Sharing System
3. Multiprogramming Operating System
4. Real Time Operating System

Types of Os -Batch Operating System

- A batch operating system is a type of operating system that manages and executes batch processing tasks without user interaction.
- In a batch processing environment, users submit jobs to the computer system in the form of batches.
- These batches contain a sequence of commands or programs to be executed, and the operating system processes them one after another without any immediate user intervention.
- Batch operating systems were prevalent in the early days of computing and are still used in some specialized applications today.



Types of Os -Batch Operating System

- **Advantages:**

Move much of the work of the operator to the computer.

Increase performance since it was possible for job to start as soon as the previous job finished.

- **Disadvantages:**

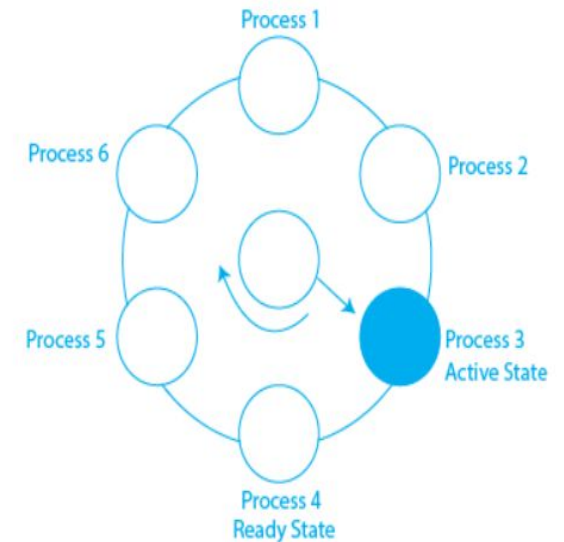
Large *Turnaround time*.

More difficult to debug program.

Due to lack of protection scheme one batch job can affect pending jobs.

Types of Os -Time Sharing System

- Multiple jobs are executed simultaneously by switching the CPU back and forth among them.
- The switching occurs so frequently (speedy) that the users cannot identify the presence of other users or programs.
- Users can interact with his program while it is running in timesharing mode.
- Processor's time is shared among multiple users. An interactive or hands on computer system provides online communication between the user and the system.
- A time shared operating system uses CPU scheduling and multiprogramming to provide each user with a small portion of a time shared computer.
- Each user has at least one separate program in memory.



Types of Os - Time Sharing System

- **Advantages :-**

Easy to use,

User friendly,

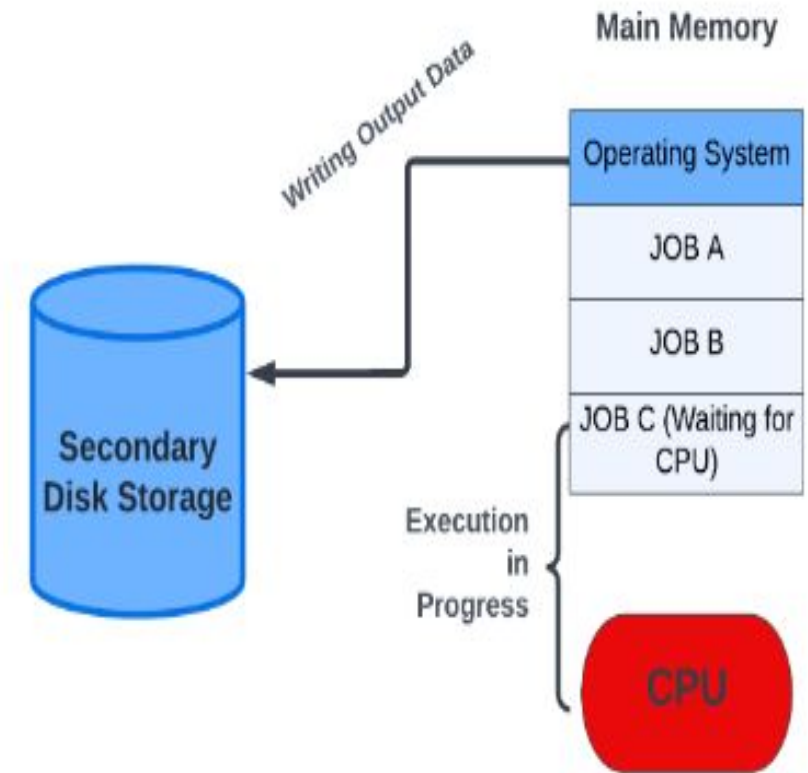
Quick response time

- **Disadvantages:-**

If any problem affects the OS, you may lose all the contents which have stored already. Unwanted user can use your own system in case if proper security options are not available

Types of Os - Multiprogramming Operating System

- Multiprogramming is technique to execute number of program simultaneously by single cpu.
- The os picks and begin to execute one of the job in main memory.
- If any input output happens in process then cpu switch to another job it is called multiprogramming
- The os allows multiple jobs to execute multiple job in single cpu.
- Advantages
 - Cpu is not ideal at any time
 - Efficient memory utilization
 - Throuput* put time is increase



Types of Os - Real Time Operating System

- A real time operating system is used, when there are rigid (strict) time requirements on the operation of a processor or the flow of data.
- It is often used as a control device in a dedicated application. Systems that control scientific experiments, medical imaging systems, and industrial control system are real time systems.
- These applications also include some home appliance system, weapon systems, and automobile engine fuel injection systems.
- Real time Operating System has well defined, fixed time constraints.
- Processing must be done within defined constraints or the system will fail. Since meeting strict deadlines is crucial in real time systems, sometimes an operating is simply a library linked in with the application programs.

Types of Os - Real Time Operating System

There are two types of real time operating system,

Hard real system:

Hard real-time systems have strict timing requirements. Tasks must be completed within a guaranteed time frame to ensure the proper functioning of critical applications. Failure to meet these deadlines can lead to catastrophic consequences.

Soft real system:

Soft real-time systems prioritize responsiveness but with some flexibility in meeting deadlines. These systems do not have stringent timing requirements and allow occasional delays without compromising the overall functionality.

Introduction of OS Process

- Process is a program under execution.
- It is an instance of an executing program, including the current values of the program counter, registers & variables.
- Process is an abstraction of a running program.
- Process is start when cpu allocate to program
- There are many resources are required to create process such as memory , cpu ,i/o files
- Once a program is in the process there are many operation done on it like delete , suspend, interrupt , create and *deadlock*.

Introduction of OS process

Process	Program
A process is program in execution.	A program is set of instructions.
A process is an active/ dynamic entity.	A program is a passive/ static entity.
A process has a limited life span. It is created when execution starts and terminated as execution is finished.	A program has a longer life span. It is stored on disk forever.
A process contains various resources like memory address, disk, printer etc... as per requirements.	A program is stored on disk in some file. It does not contain any other resource.
A process contains memory address which is called address space.	A program requires memory space on disk to store all instructions.

Process - PCB (Process Control Block):

- To implement the process model, the operating system maintains a table (an array of structures) called the process table, with one entry per process, these entries are known as Process Control Block.



Process - PCB (Process Control Block):

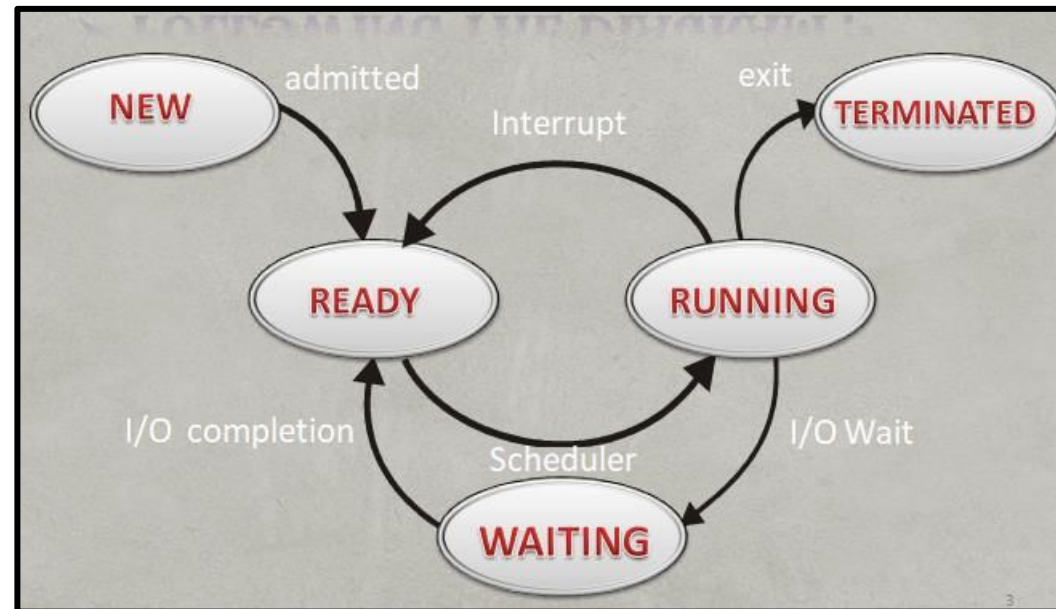
- Process Id: Each process is given Id number at the time of creation.
- Priority : it comes in number lesser number has high priority.
- Process state: The state may be ready, running, and blocked.
- Program counter: The counter indicates the address of the next instruction to be executed for this process.
- CPU registers: The registers vary in number and type, depending on the computer architecture. They include accumulators, index registers, stack pointers, and general-purpose registers, plus any condition-code information. Along with the program counter, this state information must be saved when an interrupt occurs, to allow the process to be continued correctly afterward.

Process - PCB (Process Control Block):

- CPU-scheduling(event) information: This information includes a process priority, points to scheduling queues, and any other scheduling parameters.
- Memory-management information: This information may include such information as the value of the base and limit registers, the page tables, or the segment tables, depending on the memory system used by the operating system.
- Accounting information(resource held): This information includes the amount of CPU and real time used, time limits, account numbers, job or process numbers, and soon.
- Status information(Pcb pointer): The information includes the list of I/O devices allocated to this process, a list of open files, and so on. the pcb pointer point to the next pcb in sequence

Process - Process State Transition Diagram(PSTD)

- PSTD indicate the current nature of the process it display information about the active process those are scheduled.
- When user is assign program to the system os generate PCB for that job basis on the policy process are passed in to PSTD with different five state known as process state transition diagram.

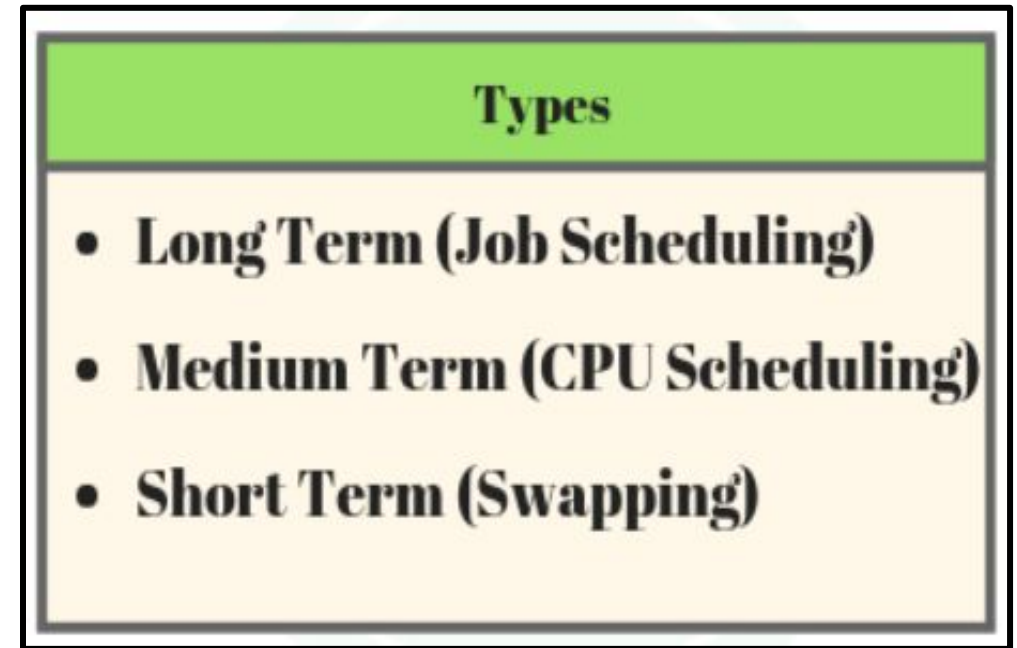


Process - Process State Transition Diagram(PSTD)

- NEW (HOLD) STATE: Any process submitted to the system it enter in new state job scheduler select the job basis on the predetrmine policy and send it to ready state.
- READY STATE: In this state process are ready to get excute , reside in main memory and wait for theprocessor.
- RUNNING STATE: Process shceduler select the job basis on the algorithm and send it to the processor there are different types of algorithm as per the os.
- WAITING STATE: During the execuation of process any input is require then it send to the waiting state and then ready state.if timer interrupt arise then it directly send to ready state.
- TERMINATED(FINISH) STATE: Process shceduler terminate the process. It may be many situation like finish the process,delete etc.

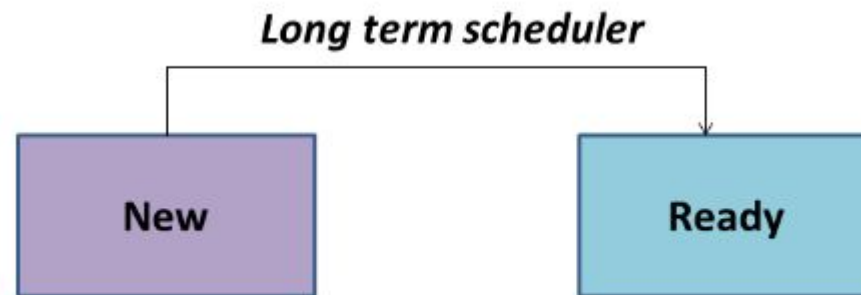
Process - Scheduler

- A program or a part of operating system that arranges jobs or a computer's operations into an appropriate sequence is called scheduler.
- Their main task is to select the jobs to be submitted into the system and to decide which process to run.
- Schedulers are of three types:
 1. Long Term Scheduler
 2. Short Term Scheduler
 3. Medium Term Scheduler



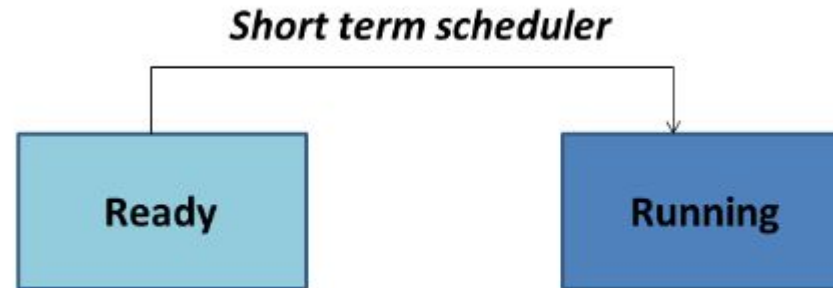
Process Scheduler - Long Term Scheduler

- Long term scheduler determines which programs are admitted to the system for processing.
- The primary objective of the job scheduler is to provide a balanced mix of jobs, such as I/O bound and processor bound.
- It also controls the *degree of multiprogramming*.
- Time-sharing operating systems have no long term scheduler.
- When process changes the state from new to ready, then there is use of long term scheduler.



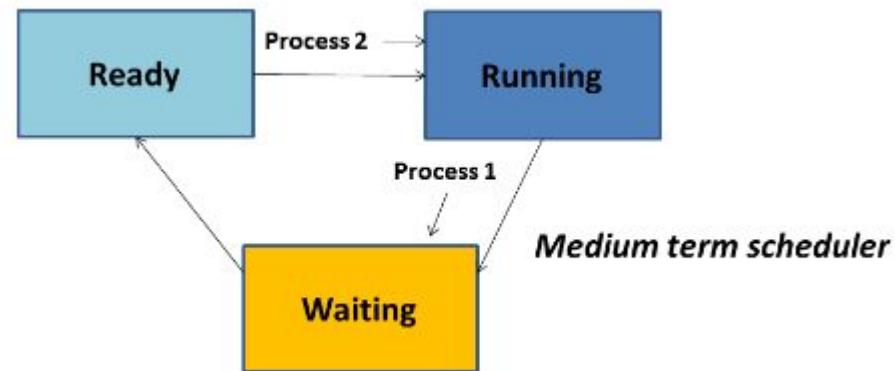
Process Scheduler - Short Term Scheduler

- It is also called CPU scheduler.
- Main objective is increasing system performance in accordance with the chosen set of criteria.
- It is the change of ready state to running state of the process.
- Short term scheduler is faster than long term scheduler.



Process Scheduler - Medium Term Scheduler

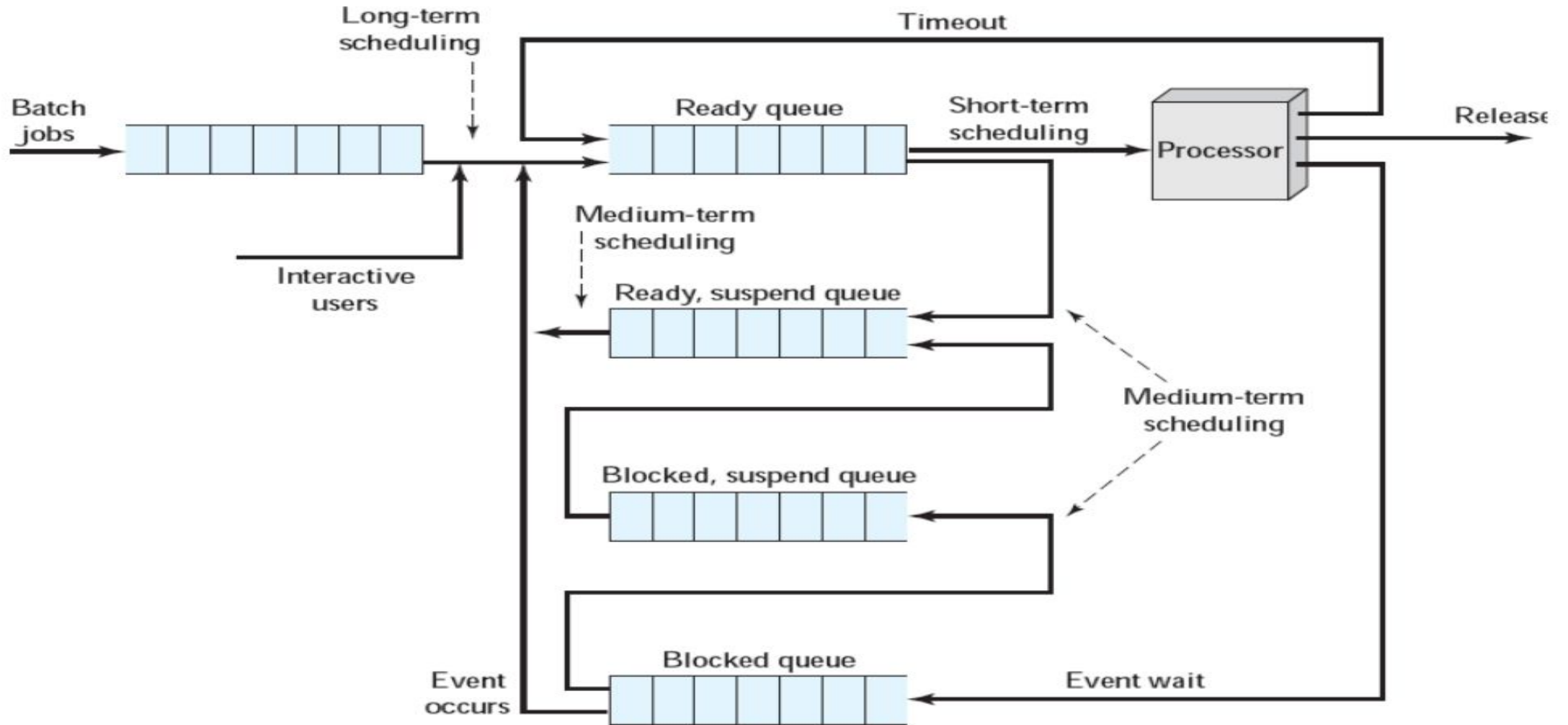
- Medium-term scheduler reduces the degree of multiprogramming.
- Medium term scheduling is part of the swapping.
- It removes the processes from the memory.
- It is a part of time sharing system.
- It swaps-out processes from main memory to secondary memory.
- Operates more frequently than long-term scheduler but less frequently than short-term scheduler.



Process Scheduler Difference

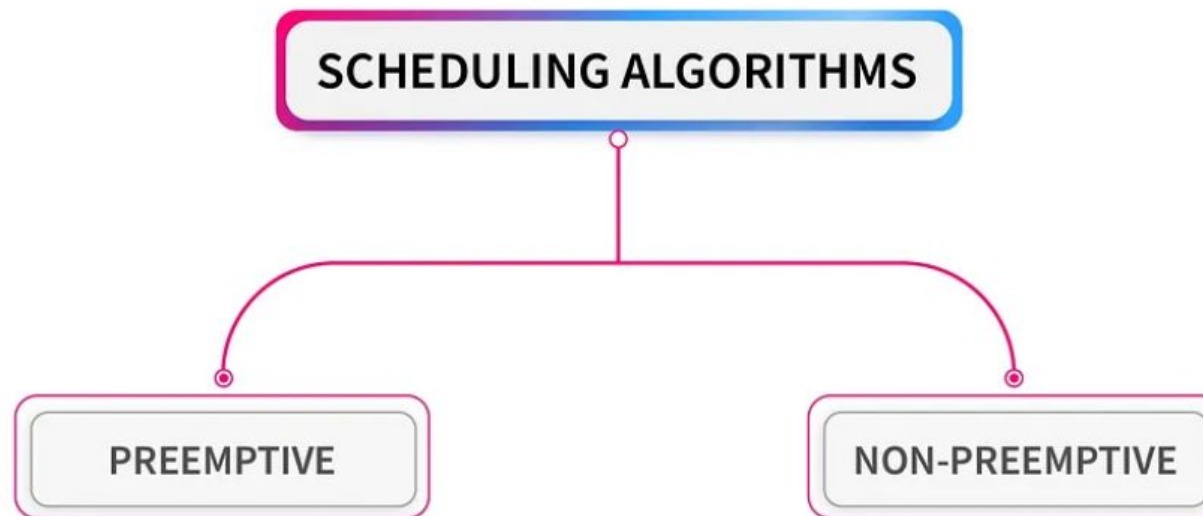
S. No.	Long-Term Scheduler	Short-Term Scheduler	Medium-Term Scheduler
1	It is a Job Scheduler	It is a CPU Scheduler	It is a process swapping scheduler
2	It takes process from the job pool	It takes process from the ready state	It takes process from running or wait/dead state
3	Its speed is lesser than short-term scheduler	It is fastest among the two other schedulers	Its speed is in between long-term and short-term
4	It controls the degree of multiprogramming	It has less control over the degree of multiprogramming	It reduces the degree of multiprogramming

A Queuing diagram representation of process scheduler



Scheduling Algorithm

- A CPU scheduling algorithm is used to determine which process will use CPU for execution, and which processes to hold or remove from execution.



Preemptive Scheduling Algorithms

- Preemptive Scheduling is a CPU scheduling technique that works by dividing time slots of CPU to a given process. The time slot given might be able to complete the whole process or might not be able to it.
- In preemptive scheduling algorithm running process may be interrupted by high priority process with low priority process.

Non-Preemptive Scheduling Algorithms

- in non-preemptive scheduling, any new process has to wait until the running process finishes its CPU cycle.
- A non preemptive scheduling it process scheduled request to completion state.
- In non –preemptive scheduling cpu transfer one job to another when first job gets complete.